Solutions

SET A			SET B			General Marking instructions:
37	54	15	26	53	20	1. Each output yields 1 mark, with a total of 9 marks.
20	30	22	8	13	23	2. A complete table yields 1 mark.
36	50	49	54	64	49	3. If the calculation / trace table is not found ⇒ 0 marks.
9+1 = 10 marks						

Questions

SetA

Following the given classes, find the outputs of the driver code below:

```
Exam one = new Exam(10, 4);
Exam two = new Exam(12);
Lab four = new Lab();
int x = four.compare(two);
```

Write your Outputs here:

37	54	15
20	30	22
36	50	49

```
class Lab {
 public static int sum = 15;
 public int cw = 12, marks = 5;
 public Lab() {
   cw += marks + sum - 30;
   marks = grade(cw + sum);
 }
 public Lab(int a) {
   cw = 6 + a + marks;
   Exam.sum -= 2;
 }
 public int grade(int sum) {
   marks += Lab.sum + Exam.sum + sum;
   Lab bonus = new Lab(4);
   return sum + bonus.marks;
 }
 public int compare(Exam e) {
   int hw = this.cw + e.marks + 7;
   e.cw = e.marks - marks + Exam.sum;
   System.out.println(cw+" "+hw+" "+sum);
   return cw + e.cw - hw;
 }
}
```

```
class Exam extends Lab {
 public static int sum = 22;
 public int marks = 7, x = -2;
 public Exam(int a) {
   this.cw = 2*a + Lab.sum;
   x = Lab.sum - (sum++);
 }
 public Exam(int b, int c) {
   super(c + b);
   marks = c + cw - marks;
   x = b + compare(this);
   Lab.sum = sum + super.marks;
   super.cw = cw - x + sum;
   System.out.println(sum+" "+x
                     +" "+marks);
 }
 public int grade(int hw) {
   Lab.sum = hw + cw + marks;
   return marks;
 }
}
```

SetB

Following the given classes, find the outputs of the driver code below:

```
Exam one = new Exam(14, 3);
Exam two = new Exam(11);
Lab four = new Lab();
int x = four.compare(two);
```

Write your Outputs here:

26	53	20
8	13	23
54	64	49

```
class Lab {
  public static int sum = 20;
  public int cw = 10, marks = 5;
  public Lab() {
    cw += marks + sum - 10;
    marks = grade(cw + sum);
  }
  public Lab(int a) {
    cw = 3 + a + marks;
    Exam.sum -= 2;
  }
  public int grade(int sum) {
    marks += Lab.sum + Exam.sum + sum;
    Lab bonus = new Lab(3);
    return sum + bonus.marks;
  }
  public int compare(Exam e) {
    int hw = this.cw + e.marks + 5;
    e.cw = e.marks - marks + Exam.sum;
    System.out.println(cw+" "+hw+" "+sum);
    return cw + e.cw - hw;
  }
}
```

```
class Exam extends Lab {
  public static int sum = 10;
  public int marks = 5, x = -5;
  public Exam(int a) {
    this.cw = 2 * a + Lab.sum;
    x = Lab.sum - (sum++);
  }
  public Exam(int b, int c) {
    super(c + b);
    marks = c + cw - marks;
    x = b + compare(this);
    Lab.sum = sum + super.marks;
    super.cw = cw - x + sum;
    System.out.println(sum+" "+x
                        +" "+marks);
  }
  public int grade(int hw) {
    Lab.sum = hw + cw + marks;
    return marks;
  }
}
```